A Reduced Variable Neighbourhood Search for the Beam Angle Optimisation Problem

Maicholl Gutierrez, Guillermo Cabrera-Guerrero, and Carolina Lagos

***Abstract*—Intensity Modulated Radiation Therapy (IMRT) is a widely used radiation therapy technique to treat cancer. The main goal in IMRT is to obtain a treatment plan that eliminates cancer cells from the tumour and, at the same time, damages as little as possible the Organs at Risk (OAR) around the tumour. To this end, we first need to seek the best possible set of beam angles, called beam angle configuration (BAC), to irradiate from. In this paper, we propose a reduced Variable Neighbourhood Search (rVNS) algorithm that explores the search space employing two different local search movements. Unlike traditional VNS algorithms, the rVNS we implement here does not need any transition rule to be implemented, as it includes both neighbourhoods moves at each iteration the rVNS we implement here does not need any transition rule to be implemented, as it includes two types of movements at each iteration. The first movement replaces each beam angle in the BAC by a $\pm 5^\circ$ beam angle, while the second movement replaces each beam angle in the BAC by a randomly chosen beam angle. We try our approach We test our approach on a set of clinical prostate cases from a hospital in Chile. Results show that the rVNS produces better results than both steepest descent and next descent local search algorithms using the same neighbourhood definitions. The rVNS is showed shown to be faster than the Local Search algorithms and quite competitive w.r.t. the obtained treatment plans.**

***Index Terms*—Reduced variable neighbourhood search, local search, intensity modulated radiation therapy, beam angle optimisation.**

I. Introduction

INTENSITY-MODULATED radiation therapy (IMRT) is one of the most widely used radiation therapy techniques. Its main objective is to obtain a *treatment plan* which allows us to completely eliminate the *tumour* (Planning Target Volume, PTV) while damaging the *organs at risk* (OAR) as little as possible. Obtaining high-quality treatment plans is a complex task from both clinical and mathematical points of view. Because of that, the IMRT problem is usually divided into three sequential sub-problems, namely Beam Angle Optimisation (BAO), Fluence Map Optimisation (FMO) and the Multi-Leaf Collimator (MLC) Sequencing. A common approach to tackling these three sub-problems is to solve the BAO problem (which in turn solves the FMO problem) and then solve the MLC sequencing problem. In this paper, we will focus on the BAO problem, i.e., the problem of finding a *beam angle configuration* (BAC) that leads to high-quality treatment plans.

Selecting the best possible BAC is considered a complex problem, mainly because of the large number of BACs that can be generated, even for relatively simple cases with few available beam angles [1]. For this reason, manually selecting the BACs in a trial-and-error process highly depends on medical doctors' experience. Thus, the automatic selection of BACs can be seen as a tool to improve the quality of treatment plans generated based on manually selected BACs.

To solve the BAO problem, we first need a quality measure to compare different BACs. Then, we say that a BAC $\mathcal{A}$ will be better than another BAC $\mathcal{B}$ if the treatment plan obtained by solving the FMO problem for $\mathcal{A}$ is better than the one we obtain when solving the FMO problem for BAC $\mathcal{B}$ [2]. Thus, solving the BAO problem also means solving the associated FMO problem for each evaluated BAC.

Since the number of BACs we can generate grows exponentially as the number of available beam angles increases, it is impossible to evaluate all of these BACs within clinically acceptable times. This situation has led authors to propose several incomplete techniques or *heuristics* in the literature. In the following, we present a brief literature review, focusing on those studies that consider incomplete search methods, as we do in this paper.

The authors in [3] (one of the first papers focused on the BAO problem) presented some mathematical models for the problem and proposed a simulated annealing algorithm to solve it. Simulated annealing variants have also been considered to solve the BAO problem in [4], reporting good results. Other heuristic algorithms have also been proposed to solve the BAO problem. For instance, in [5], the authors propose a genetic algorithm with a surrogate model, while in [6], the authors propose an Ant Colony Optimisation strategy to construct the BACs in an iterative manner. The authors of [7] propose an automated bi-level optimisation in which the step size is progressively reduced based on its decision rules.

In addition to incomplete search methods, strategies that aim to reduce the search space have also been proposed. In [8], the authors propose an angle reduction strategy to reduce the complexity of the problem based on a MIP model, which provides some information that allows the authors to reduce search space by discarding those beam angles that have the lowest scores values. The beam angles scores are computed based on the radiation delivered from beamlets of each beam angle after the

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MIP optimisation. Authors in [9] proposed a Branch-and-Prune technique considering 12, 24, and 36 beam angles. The authors in [10] propose two heuristic methods to reduce the search space. The first one is based on the geometry of the BAC that is evaluated. The authors propose to avoid angles that are either too close or in an opposite position of the current beam angles, as they behave similarly. The second heuristic is similar to the one proposed in [8]: the authors start with a BAC composed of 33 beam angles and calculate the contribution of each of them. Then, those beam angles that have a minor contribution are eliminated. The contribution of each beam angle is calculated using a MIP algorithm. Similarly, in [11], the authors seek to reduce the number of beam angles employing a scoring formula based on the mean square error of the PTV and OARs.

As previously mentioned, the BAO problem needs to solve the corresponding FMO problem. Unfortunately, solving the FMO problem is computationally expensive, and thus, some authors have proposed to use different techniques to estimate the FMO value so they do not need to spend time calculating it. For instance, the authors in [5] estimate the FMO value by using a surrogate model based on a neural network. The authors in [12] introduced a classification algorithm to obtain a final treatment using the water equivalent path length optimisation algorithm and the Euclidean distance to compare the proximity of the evaluated cases. Their classification algorithm uses the knowledge obtained from the RTP database of cancer cases treated in the Kyushu University Hospital. Estimation models have also been considered for the (related) Intensity Modulated Proton Therapy (IMPT) problem [13]. Other authors have recently proposed to use a convolution neural network to predict the beam angle selection based on patients' image processing [14], [15], [16].

Local search algorithms have also been considered to solve the BAO problem. The authors in [2] propose a mathematical method that combines an LS and an exact interior-point algorithm to solve the BAO problem. While the LS method was used to search the beam angle space, the interior point method was used to solve the associated FMO problem for each BAC. Based on the work in [2], authors in [17] introduced a *next descent* (ND) search approach which converges faster than the *Steepest Descent* (SD) in [2] without any major impairment on the objective function value. Similarly, the authors in [18] compare their previously proposed *next descent* approach to Tabu Search. The authors claim that their Tabu Search approach outperforms both the SD in [2] and the ND in [17] approaches. Although better in terms of objective function value, the Tabu Search approach is still slower than the ND approach. In [19], the authors introduce a local search method combined with a simulated annealing algorithm, which allows progressive exchange moves as the number of iterations increases.

As shown above, solving the BAO problem is extremely difficult. During the last 20 years, researchers have been working on different strategies to find high-quality BACs and their corresponding treatment plans. We can see that local search algorithms can provide very good results among these strategies. It is interesting that LS algorithms presented in the literature do not necessarily use the same neighbourhood definitions. For instance, authors in [18] compare different LS strategies using two different neighbourhood definitions. While one of the movements is defined as an *exploration* movement, the other is defined as an *exploitation* movement. However, strategies that combine different neighbourhood definitions within the same run are scarce, even though it is reasonable to consider combining two (or more) neighbourhoods. A first attempt to combine two neighbourhood definitions within the same algorithm was made in [20], where the authors applied a Variable Neighbourhood Search (VNS) algorithm based on the same neighbourhood moves defined in [18]. Authors in [21] present a more effective VNS-based algorithm called Variable Neighbourhood Descent (VND) algorithm. In their paper, they use five neighbourhood definitions. The authors report that their VND algorithm performs well in both objective function value and robustness. Unfortunately, VNS-based algorithms tend to be slow, requiring several iterations and objective function evaluations.

Unlike traditional VNS-based algorithms mentioned above, we propose a reduced Variable Neighbourhood Search (rVNS) in this paper. One distinctive feature of the rVNS algorithm is that it does not need any transition rule to be implemented to move from one neighbourhood definition to the next one, as it uses all neighbourhood definitions at each iteration. This strategy allows us to accelerate the algorithm's convergence without any significant impairment in the quality of the obtained treatment plans. To the best of our knowledge, no algorithm in the literature applied to the BAO problem combines two (or more) neighbourhood definitions within the same iteration, as the rVNS algorithm implemented in this paper does.

Then, this paper proposes a reduced Variable Neighbourhood Search (rVNS) algorithm to solve the BAO problem. The proposed rVNS uses two different neighbourhood definitions simultaneously within the same iteration. We apply our rVNS strategy to 7 clinical prostate cases. The rVNS algorithm is compared to LS and variable neighbourhood descent algorithms using the same neighbourhood definitions.

The remaining of this paper is organised as follows: In Section II we provide a brief introduction to the notation of IMRT. Particularly, we present the Beam Angle Optimisation problem and its associated Fluence Map Optimisation problem in this section. Then, in Section III, the proposed algorithm is described in-depth. In this section, we pay special attention to the neighbourhood definitions we use in our algorithms. In Section IV the experimental design is presented, and the obtained results are discussed. Finally, in Section V we discuss some conclusions of our work and depict some future research lines regarding this work.

II. IMRT: General Notions

In IMRT, each beam angle $\mathcal{A}_b \in \mathcal{A}$ with $b = 1, \ldots, N$ and N the total number of beam angles in the BAC $\mathcal{A}$, is discretised into *beamlets*. The vector $x \in \mathbb{R}^n$, represents the treatment plan to be optimised and its elements $x_i \in x$ represent the amount of radiation that will be delivered by the i-th beamlet of the BAC $\mathcal{A}$. Using the MLC device, we are able to modulate the radiation that is delivered by each beamlet. In the same way, each organ each structure is discretised into sub-volumes called *voxels*. The amount of radiation or *dose* deposited at each voxel j of the region r (tumour or organ at risk) by treatment plan x is denoted by $d_j^r(x)$. To calculate $d_j^r(x)$ the following equation is used:

$$d_j^r(x) = \sum_{i=1}^{n} A_{ji}^r x_i \forall j = 1, 2, \ldots, m^r, \tag{1}$$

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where A^r_{ji} is the intensity matrix and defines the rate at which radiation from the i-th beamlet is deposited in the j-th voxel of the region r with $r \in R = O_1, \ldots, O_Q, T$ where the tumour is indexed by $r = T$ and the organs at risk by $r = O_q$ with $q = 1, \ldots, Q$. The parameter m^r corresponds to the number of voxels in the region r. For more information on the notation used in this paper, refer to the work in [2].

As mentioned in Section I, to solve the BAO problem, we need to solve the associated FMO problem for each evaluated BAC. Then, in the next two sections, we introduce the FMO problem and, then, the BAO problem.

A. Fluence Map Optimisation

The FMO problem can be modeled as follows: Let $z(x) : \mathcal{X}(\mathcal{A}) \to \mathbb{R}^*_{\geqq 0} := \{\upsilon \in \mathbb{R} : \upsilon \geq 0\} \cup \{\infty\}$ the objective function of the FMO problem. The goal is to minimise $z(x)$ for a particular BAC $\mathcal{A}$. As mentioned in the previous section, the objective function we use in this work is based on the logistic function [22]. This function is mainly based on the generalised equivalent uniform dose (*gEUD*). The *gEUD* was originally proposed by [23], and it is defined as the biologically equivalent dose that, irradiated uniformly, would produce the same response as the non-uniform dose actually delivered to the patients. The mathematical expression for the *gEUD* is presented in (2).

$$gEUD^r(x) = \left(\frac{1}{m^r} \sum_{j=1}^{m^r} \left(d^r_j\right)^{a^r} \right)^{\frac{1}{a^r}}, \tag{2}$$

Parameter a^r is associated to the r-th region. For the tumour, we make $a^r < 0$, and for OARs we make $a^r > 1$. When $|a^r|$ increases (decreases), the gEUD becomes more sensitive to less (more) irradiated voxels in the tumour (OARs). Therefore, for those OARs that allow certain levels of radiation without a functional compromise (also called parallel organs), the parameter a^r is set close to 1. Making use of the *gEUD* expression, we can introduce the logistic function as follows:

$$\begin{aligned} \min_{x \in X(\mathcal{A})} \quad & z_{log}(x) = -\sum_{q=1}^{Q} \ln U\left(x; O_q, v_{O_q}, eud_0^{O_q}\right) \\ \text{s.t.} \quad & gEUD^T(x) \geq eud_0^T \end{aligned} \tag{3}$$

where

$$U\left(x; O_q, \nu_{O_q}, eud_0^{O_q}\right) = \left(1 + \left(\frac{gEUD^{O_q}(x)}{eud_0^{O_q}}\right)^{\nu_{O_q}}\right)^{-1}. \tag{4}$$

The parameters eud_0^T and $eud_0^{O_q}$ correspond to the *gEUD* value prescribed for the tumour and organs at risk, respectively. The parameter $\nu_{O_q} > 0$ represents the importance of the q-th organ at risk and $gEUD^r$ corresponds to the value of the *gEUD* of the region r. Similar objective functions have been proposed in the literature before (see e.g. [2], [24], [25]). As mentioned in [2], when the parameters of the logistic function are conveniently set, the function is convex.

B. Beam Angle Optimisation

The BAO problem is an NP-Hard combinatorial optimisation problem, in which we must find a combination of N beam angles among all possible combinations. As mentioned, to compare between two BACs, we need to find the best possible treatment plan we can generate using such BACs, i.e. we need to solve the FMO problem for each BAC we are evaluating. Then, we can say that depending on the type of objective function considered for the FMO, the integrated BAO+FMO problem we solve in this paper can be a mixed-integer optimisation problem, a mixed integer non-linear optimisation problem, or an integer programming problem.

Since we use an objective function z that is non-linear, we shall say that the problem to be solved in this paper is a mixed-integer non-linear problem. Also, for this work, we do not consider all available co-planar beam angles, but instead, we consider those angles separated by 5°starting from the beam angle 0°, i.e. $\mathcal{P}$ ={0°, 5°, 10°,..., 345°, 350°, 355°}. Thus, we have 72 available beam angles, from which we must choose only $N = 5$ beams to get a BAC. Then, we say that we have $\mathcal{P}^N = \binom{\mathcal{P}}{N}$ available BACs. Since evaluating each BAC take several seconds, it is not possible for us to enumerate all these available BACs in a clinically acceptable time. Then, we are looking for a way to solve the BAO problem with some incomplete technique that allows us to obtain good quality treatment plans within a reasonable time. Mathematically, we can express the BAO problem as:

$$\begin{aligned} \min \quad & h(\mathcal{A}) \\ \text{s.t.} \quad & \mathcal{A} \in \mathcal{P}^{\mathcal{N}}, \end{aligned} \tag{5}$$

where $h(\mathcal{A})$ is the optimal value of the FMO problem, FMO($\mathcal{A}$) introduced in the previous section and $\mathcal{P}^N$ is, again, the set of all the available BACs that consider $N = 5$ (different) beam angles from set $\mathcal{P}$. We propose solve solving the BAO Problem through a VNS-based algorithm.

III. Reduced Variable Neighbourhood Search (rVNS)

The Variable neighbourhood Search algorithm is a metaheuristic that works exceptionally well in discrete spaces [26]. The proposed reduced Variable Neighbourhood Search (rVNS) algorithm is a VNS-based algorithm that simultaneously uses two different neighbourhood definitions within the same iteration. The rVNS is a variant of the VNS framework that has been shown to perform very well for large combinatorial problems [27] such as the one we address in this paper. One distinctive feature of rVNS is that it does not need any transition rule to move from one neighbourhood to the other. Unlike traditional VNS algorithms, where only one neighbourhood definition is used to generate the neighbours at each iteration, the rVNS generates the neighbours using all the available neighbourhood definitions at each iteration. Then, the rVNS algorithm "explores" and "exploits" the search space simultaneously at each iteration. This characteristic makes the rVNS algorithm a simple yet efficient strategy, as no transition rule must be implemented. Also, the stopping criterion is similar to the one used in local search algorithms, and the algorithm stops when no neighbour is better than the current solution.

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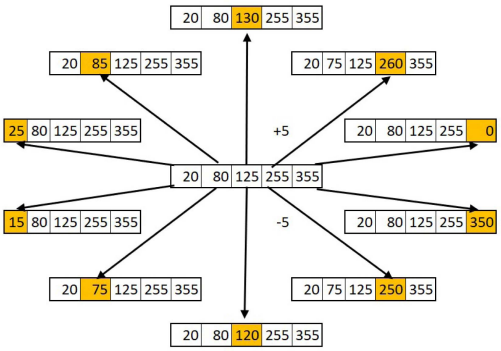

Fig. 1. Neighbourhood definition $\mathcal{N}_1$.

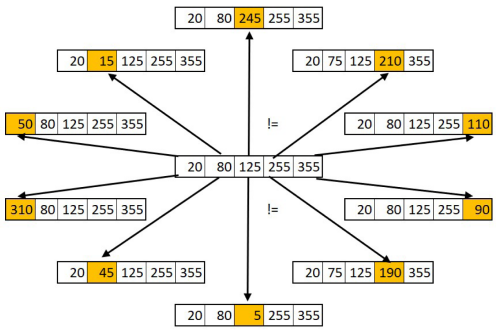

Fig. 2. Neighbourhood definition $\mathcal{N}_2$.

Before we start explaining the rVNS algorithms we implemented in this paper, we need to introduce the neighbourhood definitions we are going to use within our algorithms.

A. Neighbourhood Definitions

The first neighbourhood definition we consider in this paper is the one previously introduced in [2], [28]. This move, we call $\mathcal{N}_1$, consists of a variation of $\pm 5^\circ$ at each beam angle $\mathcal{A}_i$. This movement generates a neighbourhood consisting of $2 \times N$ BACs, with N the number of beam angles in a BAC. As shown in Fig. 1, for our case, the neighbourhood size would be 10. Note that, for this neighbourhood definition, we can generate the entire neighbourhood of the current solution at each iteration, as the number of neighbours of a solution is a linear function of N.

The second neighbourhood definition we consider in this paper is the one introduced in [18]. This move, we call $\mathcal{N}_2$, consists in replacing one beam angle $\mathcal{A}_i$ in the BAC $\mathcal{A}$ with another available beam angle that does not belong to $\mathcal{A}$. Since the number of available angles is very large, we cannot generate the entire neighbourhood at each iteration as it consists of $(72 - N) \times N$ possible BACs. Thus, in our case, for each beam angle in $\mathcal{A}$, we randomly choose only two available beam angles that do not belong to $\mathcal{A}$. As shown in Fig. 2, this movement generates a neighbourhood consisting of $2 \times N$ BACs, with N, again, the number of beam angles in BAC $\mathcal{A}$.

Based on the neighbourhood definitions above, we can easily note that both strategies surf the search space in a different manner. While neighbourhood definition $\mathcal{N}_1$ favours the exploitation of the search space that is nearby the current BAC, neighbourhood definition $\mathcal{N}_2$ favours the exploration of the search space by performing "bigger" steps. Having these features in mind is important to design a VNS-based algorithm that will somehow combine these two movements. It is also important to know that the complexity of our rVNS algorithm (and for LS algorithms in general) depends mainly on the size of the generated neighbourhood. In our case, this is polynomial to the value of N.

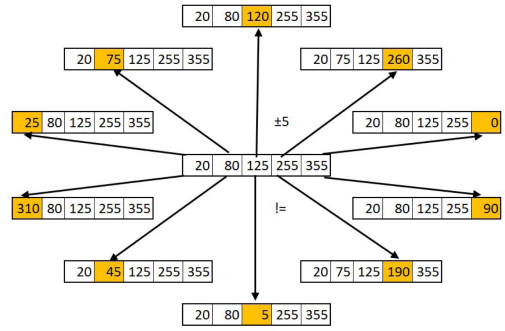

Fig. 3. Example of a neighbourhood generated by the rVNS algorithm using both the $\mathcal{N}_1$ and the $\mathcal{N}_2$ neighbourhood definitions.

B. Reduced Variable Neighbourhood Search Algorithm

The proposed rVNS generates a neighbourhood using all the available neighbourhood definitions. In our case, we used two definitions for the BAO problem, explained in the Section III-A. Unlike LS algorithms, in VNS-based algorithms, we do not need any specific input parameter regarding neighbourhood definitions. That is because the generated neighbourhood consists of solutions generated by using $\mathcal{N}_1$ and $\mathcal{N}_2$, as we show in Fig. 3. Since we have 5 beams per BAC, five neighbours are generated using the $\mathcal{N}_1$ neighbourhood definition, while the other five neighbours are generated using neighbourhood definition $\mathcal{N}_2$. It is important to note that both the perturbed angle and the neighbourhood definition used to replace such beam angle, are randomly chosen.

The rVNS proposed in this paper to solve the BAO problem is presented in Algorithm 1. From line 1 to line 5, the initial solution is generated and set as the current solution (incumbent). Note that the initial BAC can be either received as input from medical doctors or randomly generated. After setting up the current solution $\mathcal{A}$, in line 7 the neighbourhood of $\mathcal{A}$ is generated. The function in line 10 **generateNextNeighbourhood**($\mathcal{B}$, $\mathcal{A}$,**"SD"**) is a key part of our algorithm and, therefore, we devote an entire section (see Section III-B1) to explain in detail how it works. Then, in line 10, the algorithm compares the best neighbour, $\mathcal{A}'$, to the current solution $\mathcal{A}$. The algorithm stops if the current solution is better than the best neighbour. The algorithm updates the current solution with the best neighbour, otherwise.

As mentioned, the **generateNextNeighbourhood** function in Algorithm 1, will be explained in the next section.

1) Generate Next Neighbourhood Function: The **generateNextNeighbourhood** function in Algorithm 1 needs particular attention as it is a key part of our approach. This function

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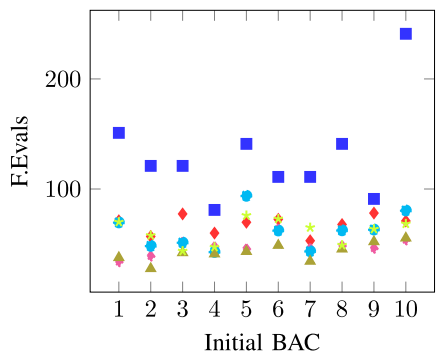
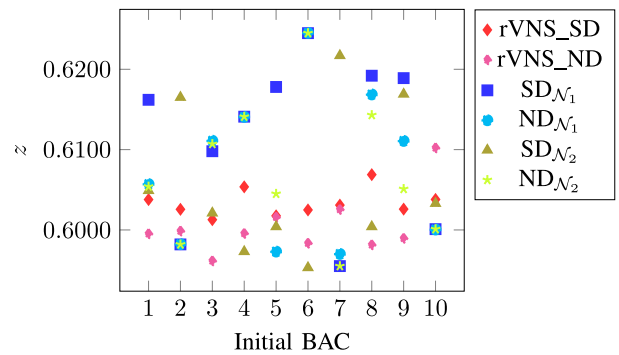
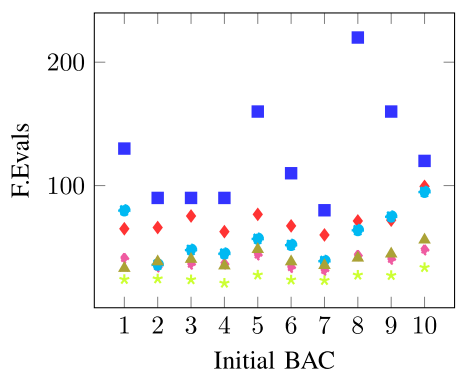


(a) Function evaluations per initial BAC for the CERR case.

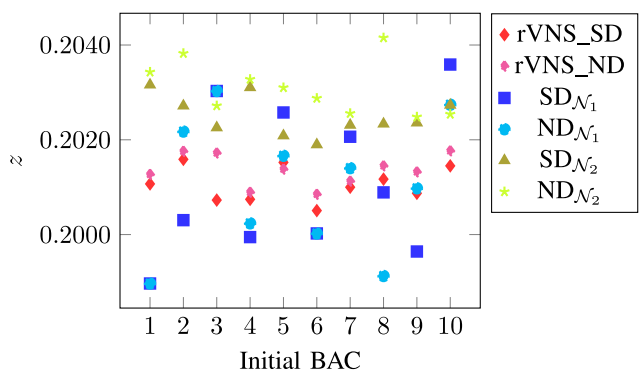


(b) Average objective function values per initial BAC for the CERR case.

Fig. 4. Results obtained for each algorithm per initial BAC for the CERR instance



(a) Function evaluations per initial BAC for the TRT004 case



(b) Average objective function values per initial BAC for the TRT004 case.

Fig. 5. Results obtained for each algorithm per initial BAC for the TRT004 instance

allows us to find the best neighbour of a BAC, for a given (set of) neighbourhood definition(s). This function needs the current solution $\mathcal{A}$ and a set $\mathcal{B}$ consisting of one or more neighbourhood definitions $\mathcal{B} = \{\mathcal{N}_1, \ldots, \mathcal{N}_k\}$. For each beam angle in BAC $\mathcal{A}$, we randomly choose a neighbourhood definition from $\mathcal{B}$ and call function **generateNeighbour**, which will return a neighbour of $\mathcal{A}'$under the selected neighbourhood definition. If the obtained treatment plan is better than the best neighbour found so far by the **generateNextNeighbourhood** function, the best neighbour is overwritten. It is important to note that we make sure that the number of neighbours generated under each neighbourhood definition in $\mathcal{B}$ is uniformly distributed using the Function **randPossibleNeighbour**. An example of how the two available neighbourhood definitions, $\mathcal{N}_1$ and $\mathcal{N}_2$ are distributed when generating a neighbourhood is shown in Fig. 3. In addition, the **generateNextNeighbourhood** function requires a third input parameter which allows it to define the search strategy. The allowed values for this parameter are "SD" and "ND," which stand for *steepest descent* and *next descent* strategies, respectively. If the input parameter is "SD," then the **generateNextNeighbourhood** function explores the entire neighbourhood before returning the best neighbour. Unlike "SD," if the "ND" strategy is selected, then the first neighbour with a better objective function value than the current solution is returned. Note that, for this case, we don't necessarily explore the entire neighbourhood of the current solution. Thus, the **generateNextNeighbourhood** function returns the neighbour with the best objective function value in the SD case or the first solution found with a better objective function value than the current solution in the ND case.

We use the **generateNeighbour** function in Algorithm 2 to generate a neighbour of a BAC for a given beam angle and neighbourhood definitions. Using this function in an iterative manner, allows the rVNS algorithm to generate the entire neighbourhood of a particular BAC. We then say that **generateNeighbour** $((\mathcal{N}_n, \mathcal{A}, b))$ function generates the neighbour of BAC $\mathcal{A}$ where the b-th beam angle of BAC $\mathcal{A}$ is replaced according to the neighbourhood definition $\mathcal{N}_n$.

C. Benchmark Algorithms

As mentioned in previous sections, we compare our rVNS algorithm to well-known LS algorithms from the literature, namely Steepest Descent (SD) [2], [17], [18] and Next Descent

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Algorithm 1: Reduced VNS Algorithm.

Input: N (# beam angles in a BAC), $\mathcal{A}^0$ (initial BAC, if any), $[\mathcal{N}_1, \ldots, \mathcal{N}_k]$ (neighbourhood definitions)
Output: $\mathcal{A}$ (locally optimal BAC)
Begin:
if $\mathcal{A}^0 = \emptyset$ **then**
  $\mathcal{A}^0$ =generateInitSol($\mathcal{P}^N(K)$) ;
  $\mathcal{B}$ = $[\mathcal{N}_1, \ldots, \mathcal{N}_k]$;
end
$\mathcal{A} = \mathcal{A}^0$;
repeat
  stop = true;
  $\mathcal{A}^{'}$ = generateNextNeighbourhood($\mathcal{B}, \mathcal{A}$, "SD");
  if $h(\mathcal{A}^{'}) > h(\mathcal{A})$ **then**
    stop = false;
  else
    $\mathcal{A} = \mathcal{A}^{'}$;
  end
until stop;
return $(\mathcal{A}^{'})$;

Algorithm 2. generateNextNeighbourhood.

Input: $[\mathcal{N}_1, \ldots, \mathcal{N}_k]$ (neighbourhood definitions), $\mathcal{A}$ (Current Solution), $\mathcal{O}$ (method of neighbourhood solution)
Output: $\mathcal{A}$ (Next Neighbour BAC)
Initialisation:
$\mathcal{B}$ = $[\mathcal{N}_1, \ldots, \mathcal{N}_k]$;
$\mathcal{I}$ = $[1, 1, 2, 2, \ldots, N, N]$;
repeat
  n = randPossibleNeighbour($\mathcal{B}$);
  b = rand($\mathcal{I}$);
  $\mathcal{A}^{'}$ =generateNeighbour($\mathcal{N}_n, \mathcal{A}, b$);
  $\mathcal{I} = \mathcal{I} \setminus \{b\}$;
  if $h(\mathcal{A}^{'}) < h(\mathcal{A})$ **then**
    $\mathcal{A} = \mathcal{A}^{'}$;
    if $\mathcal{O}$ == *"ND"* **then**
      break;
    end
  end
until $\mathcal{I} = \emptyset$;
return $\mathcal{A}$;

(ND) [17], [18]. We set some of the parameters of the rVNS algorithm, such as the neighbourhood size, equal to those from the literature, so a fair comparison can be made. The neighbourhood size and the number of function evaluations from the different algorithms are the same for each iteration. The remaining of this section is devoted to briefly explaining each of these local search algorithms.

Algorithm 3. Steepest Descent Algorithm.

1 **Input:** N (# beam angles in a BAC), $\mathcal{A}^0$ (initial BAC, if any), $\mathcal{B} = [\mathcal{N}_x]$ (neighbourhood def.)
Output: $\mathcal{A}$ (locally optimal BAC)
2 **Begin:**
3 **if** $\mathcal{A}^0 = \emptyset$ **then**
4 $\mathcal{A}^0$ = generateInitSol($\mathcal{P}^N(K)$) ;
5 **end**
6 $\mathcal{A} = \mathcal{A}^0$;
7 **repeat**
8 stop = true;
9 $\mathcal{A}^{'}$ = generateNextNeighbourhood($\mathcal{B}, \mathcal{A}$,"SD");
10 **if** $h(\mathcal{A}^{'}) > h(\mathcal{A})$ **then**
11 stop = false;
12 **else**
13 $\mathcal{A} = \mathcal{A}^{'}$;
14 **end**
15 **until** stop;
16 **return** $(\mathcal{A})$;

1) Local Search Strategies: In the SD strategy, the entire neighbourhood of the current solution is generated at each iteration. The current solution is updated iff the best neighbour is better than the current solution. Otherwise, the algorithm stops. We then consider the SD twice: $\text{SD}_{\mathcal{N}_1}$ and $\text{SD}_{\mathcal{N}_2}$, where the first one is an SD algorithm considering the $\mathcal{N}_1$ neighbourhood definition, and the second one is an SD algorithm that considers the $\mathcal{N}_2$ neighbourhood definition. Note that the main difference with the rVNS algorithm proposed in this paper is that the set $\mathcal{B}$ passed on to the function **generateNextNeighbourhood**($\mathcal{B}$, $\mathcal{A}$,$''$**SD**$''$) consists of only one neighbourhood definition. Algorithm 3 shows the pseudo-code of the implemented SD. From line 1 to line 5, the initial solution is generated and set as the current solution (incumbent). Note that the initial BAC can be either received as input from medical doctors or randomly generated. After setting up the current solution $\mathcal{A}$, in line 6, the neighbourhood of $\mathcal{A}$ is generated by using function **generateNextNeighbourhood**($\mathcal{B}, \mathcal{A}$,**"SD"**), with $\mathcal{B} = \{\mathcal{N}_n\}$(line 9). Then, the algorithm compares the best neighbour, $\mathcal{A}'$, to the current solution $\mathcal{A}$ in line 10. The algorithm stops if the current solution is better than the best neighbour. The algorithm updates the current solution with the best neighbour, otherwise.

Unlike the SD strategy, in the ND strategy we do not necessarily need to generate the entire neighbourhood. In the ND algorithm, each neighbour is generated at a time following a random order. Every time a neighbour is generated, we evaluate it against our current solution. If the generated neighbour is better than the current solution, we stop the neighbourhood generation and update the current solution. Then, while the SD always generates (for our case) 10 neighbours at each iteration, the ND might generate from only one neighbour (in case the first generated neighbour is better than the current solution) to 10 neighbours per iteration. Also, note that, in the SD algorithm, the

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Algorithm 4. Next Descent Algorithm.

```
Input: N (# beam angles in a BAC), A^0
       (initial BAC, if any), B = [N_x]
       (neighbourhood def.)
Output: A (locally optimal BAC)
Begin:
if A^0 = ∅ then
    A^0 = generateInitSol(P^N(K)) ;
end
A = A^0 ;
repeat
    stop = true;
    A' = generateNextNeighbourhood(B,
     A, "ND");
    if h(A') > h(A) then
        stop = false;
    else
        A = A';
    end
until stop;
return (A);
```

order in which the neighbours are generated is not relevant. For the ND strategy, the order in which the neighbours are generated is important, as might be some neighbours that are better than the current solution (and even the best one in the neighbourhood) that will be simply not generated because another neighbour, better than the current solution, was generated before.

Similarly to what we do for the SD algorithm, we consider the ND algorithm twice: ND $_{\mathcal{N}_1}$ and ND$_{\mathcal{N}_2}$. For more details on these algorithms and their application in IMRT, see [2], [17], [18]).

We will test our rVNS algorithm on a set of 6 clinical prostate cases from a Hospital in Chile. The results will be compared to both the SD and ND local search strategies previously proposed in the literature.

IV. Experiments

We present in this section the computational experiments carried out in this work. We use an Intel i7 processor with 2.6 GHz cores and 32 GB of RAM to run our experiments. Windows 10 was the operating system. Every algorithm in this paper was coded in JAVA 17 language using NetBeans IDE. Experiments were performed on a prostate case obtained from the Computational Environment for Radiotherapy Research [29], and 6 prostate cases from a Hospital in Chile, called TRT003, TRT004, TRT007, TRT008, TRT009 and TRT010. For all these cases, the tumour is located at the prostate while the considered OARs are the bladder and the rectum. Note that the boundaries of the tumour and the bladder are overlapped which makes the problem a very complex one [18], [30].

In the CERR prostate case, the tumour has more than 7,000 voxels, the rectum about 5,500, and the bladder around 9,500. Although the same multileaf collimator is used for all beam angles, the number of beamlets considered in the optimisation process depends on each beam angle. This is because those beamlets that do not irradiate at least one voxel of the tumour are not considered decision variables in the optimisation process. Thus, the number of decision variables (beamlets) depends on the BAC and ranges between 160 and 220. Instances TRT00X present similar numbers of voxels and beamlets. The number of beam angles $\mathcal{N}$ considered in a BAC is equal to 5 for all instances. We use Knitro Artelys [31] as the solver for all nonlinear optimisation problems.

We run each algorithm 30 times per instance, as accepted value for statistical analysis [32], and focus on determining whether the data of our benchmarks follow a normal distribution. To this end, we perform a Kolgomorov-Smirnov-Lilliefors's test [33] assuming the hypothesis H_0: data follow a normal distribution and H_1: otherwise. We obtain p-values lower than 0.05 for all six cases. Hence, we cannot assume that the data follow a normal distribution. As normality cannot be assumed and samples are independent, we then apply nonparametric Wilcoxon-Mann-Whitney's test [34] to compare our algorithms. We study whether there is a significant difference in the average values obtained by each algorithm. We use the Wilcoxon-Mann-Whitney's test [34] with the hypothesis $H_0 : \eta_1 - \eta_2 \leq 0$ and H1 : $\eta_1 - \eta_2 > 0$. Where η_1 and η_2 are the average values. For each instance, we performed 30 pairwise comparisons for each initial BAC for SD$_{\mathcal{N}_1}$, ND$_{\mathcal{N}_1}$, SD$_{\mathcal{N}_2}$, ND$_{\mathcal{N}_2}$, rVNS_SD and rVNS_ND algorithms to analyse the obtained p-values considering a significance level of 0.05. Here η_1 are column algorithms, and η_2 are row algorithms. We focus on the rVNS_SD and rVNS_ND where we reject the null hypothesis as the p-value is less than the significance level. Table V in the Appendix section shows a summary of the results obtained after applying the Wilcoxon-Mann-Whitney's test.

Table I(a) shows the values of the parameters used in the experiments; these were obtained in the literature in [2]. Table I(b) shows the 10 initial BACs considered in the experiments. These BACs were randomly generated, ensuring a difference of, at least, 10° between beam angles. Finally, Table I(c) presents the optimal values for the FMO problem obtained by each initial BAC at each case.

We need to point out that the SD algorithm generates the entire neighbourhood at each iteration, and then we can consider it as a deterministic algorithm (i.e. running the algorithm from starting BAC $\mathcal{A}$ will always lead us to the same locally optimal BAC $\mathcal{A}^*$). Then, we only run this algorithm once for each initial BAC from Table I(b). Unlike the SD strategy, the ND and the rVNS strategy are considered stochastic ones, and then 30 runs per initial BAC must be performed for statistical analysis purposes.

We will highlight this study's main results, focusing on the CERR and the TRT004 cases. More details on the remaining cases' results can be found in the Appendix section.

Fig. 4(a) graphically presents the number of function evaluations through the FMO for each initial BAC. As we can see, the algorithm that needs more function evaluations is the SD$_{\mathcal{N}_1}$, with an average of 131 per initial BAC. This number drops significantly to 42.8 when the same SD strategy is used considering the $\mathcal{N}_2$ neighbourhood definition. Unlike this, for the ND strategy, the number of function evaluations remains stable at around 61 for both $\mathcal{N}_1$ and $\mathcal{N}_2$ neighbourhood definitions. As expected, the

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TABLE I
PARAMETERS USED IN THE EXPERIMENTS AND THE OPTIMAL FMO VALUES FOR EACH INSTANCE

(a) Parameters used for FMO

r	eud_0^r	a^r	v_r
Tumor	76	-10	12
Rectum	60	8	8
Bladder	65	2	5

(b) Initial BACs

BAC	$\mathcal{A}_1$	$\mathcal{A}_2$	$\mathcal{A}_3$	$\mathcal{A}_4$	$\mathcal{A}_5$
1	5	105	185	230	310
2	60	110	210	255	335
3	60	140	155	255	355
4	15	100	160	240	335
5	45	145	210	220	325
6	5	40	230	275	340
7	95	210	245	270	340
8	30	120	130	170	345
9	80	155	265	270	335
10	20	125	185	220	305

(c) Optimal FMO Values for the Initial BACs

BAC	CERR	TRT003	TRT004	TRT007	TRT008	TRT009	TRT010
1	0.6303	0.2383	0.2101	0.3499	0.2183	0.2602	0.5983
2	0.6198	0.2257	0.2095	0.2853	0.2138	0.2561	0.5840
3	0.6455	0.2290	0.2097	0.3679	0.2209	0.2656	0.6032
4	0.6295	0.2357	0.2105	0.3388	0.2182	0.2608	0.5900
5	0.7277	0.2715	0.2322	0.7679	0.2406	0.2770	0.6539
6	0.6399	0.2380	0.2124	0.3513	0.2250	0.2647	0.6093
7	0.6230	0.2254	0.2041	0.2817	0.2134	0.2584	0.5876
8	0.6865	0.2726	0.2090	0.4916	0.2262	0.2766	0.6319
9	0.6349	0.2311	0.2159	0.3143	0.2160	0.2659	0.5865
10	0.6780	0.2530	0.2270	0.4249	0.2274	0.2743	0.6470

TABLE II
RESULTS OBTAINED FOR EACH ALGORITHM AND INITIAL BAC, FOR THE CERR INSTANCE

	rVNS_SD			rVNS_ND			$\text{SD}_{\mathcal{N}_1}$			$\text{ND}_{\mathcal{N}_1}$			$\text{SD}_{\mathcal{N}_2}$			$\text{ND}_{\mathcal{N}_2}$		
BAC	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z
1	9.13	71.20	**0.6038**	10.60	**33.60**	**0.5995**	15.00	151.00	0.6162	20.50	69.67	0.6057	3.67	**37.67**	0.6049	19.00	70.00	0.6054
2	7.67	56.90	0.6026	9.4	**38.72**	0.5998	12.00	121.00	**0.5982**	14.33	48.33	**0.5982**	3.11	**27.67**	0.6165	21.60	57.80	**0.5982**
3	9.73	77.40	**0.6013**	13.68	51.53	**0.5961**	12.00	121.00	0.6098	15.33	51.33	0.6111	4.11	**42.11**	0.6021	20.40	**43.60**	0.6107
4	7.96	60.00	0.6054	12.55	47.40	**0.5995**	8.00	81.00	0.6141	10.33	**42.83**	0.6141	4.17	**41.00**	**0.5973**	21.60	47.20	0.6141
5	8.96	69.80	0.6018	12.95	**45.43**	0.6015	14.00	141.00	0.6178	30.50	93.83	**0.5973**	4.22	**43.22**	**0.6004**	21.00	76.00	0.6045
6	9.27	72.50	0.6025	17.00	63.60	**0.5983**	11.00	111.00	0.6245	15.33	**62.33**	0.6245	4.78	**48.78**	**0.5953**	17.00	72.80	0.6245
7	7.27	53.00	0.6031	12.80	45.80	0.6025	11.00	111.00	**0.5955**	12.17	**43.50**	0.5970	3.33	**34.33**	0.6217	20.00	65.00	**0.5955**
8	8.77	67.70	0.6069	11.80	**48.20**	**0.5981**	14.00	141.00	0.6192	20.17	62.50	0.6169	4.44	**45.44**	**0.6004**	22.00	49.00	0.6143
9	9.80	78.10	**0.6026**	12.00	**45.80**	**0.5989**	9.00	91.00	0.6189	16.60	63.20	0.6111	5.11	**52.11**	0.6169	18.80	63.60	0.6051
10	9.07	71.00	0.6038	15.20	**53.40**	0.6102	24.00	241.00	**0.6001**	24.50	80.50	**0.6001**	5.44	**55.44**	0.6033	20.20	68.40	**0.6001**
$\bar{\text{x}}$	**8.76**	**67.75**	**0.6034**	**12.80**	**47.35**	**0.6005**	**13.00**	**131.00**	**0.6114**	**17.98**	**61.80**	**0.6076**	**4.24**	**42.78**	**0.6059**	**20.16**	**61.34**	**0.6072**

rVNS strategy does not need too many function evaluations to converge, although it is still slower than the $\text{SD}_{\mathcal{N}_2}$ algorithm. The rVNS strategy performs significantly faster when considering the ND strategy over the SD strategy (42.8 and 61.3 per initial BAC, respectively).

Fig. 4(b) graphically presents the obtained objective function value, z, for each algorithm at each initial BAC, for the CERR case. We need to note that the standard deviation of the rVNS_SD and the rVNS_ND (0.0017 and 0.0039, respectively) are significantly smaller than the standard deviation obtained by the LS algorithms ($\text{SD}_{\mathcal{N}_1}$: 0.0101, $\text{SD}_{\mathcal{N}_2}$: 0.0091, $\text{ND}_{\mathcal{N}_1}$: 0.0095, and $\text{ND}_{\mathcal{N}_2}$: 0.0088). In addition, we observe that the proposed rVNS algorithm presents good quality results, which are, on average(column $\bar{x}$ in Table II), better than all the other LS algorithms. It is important to note that between both rVNS algorithms, the rVNS_ND version generally presents a better z value than the rVNS_SD version.

Table II shows the initial BAC (BAC), the number of iterations the algorithm needs before convergence (Iter.), the number of function evaluations (#Evals), and the best objective function (z) value obtained by the algorithm. All these values are getting after 30 runs for every initial BACs. Thus, the reported values are the average of these 30 runs. The two algorithms with the best average performance in #Evals and z value have been highlighted. The best z value is highlighted and underlined, while the second best z value is highlighted only. We can note that the exploitation strategy ($\mathcal{N}_1$) generally presents more evaluations than the exploration strategy ($\mathcal{N}_2$).

We can also note that the rVNS_ND proposed here presents a competitive performance in both #Evals and z, obtaining the best performance among all algorithms in 4 out of 10 initial BACs. The rVNS_SD and the rVNS_ND have the most highlighted z values, with 4 and 3 out of 10 initial BACs highlighted, respectively. The rVNS algorithms obtain better median values than the LS strategies, while the rVNS_{SD} algorithm performs better than the LS strategies regarding z values.

We now present the analysis of the results obtained for the TRT004 case. Fig. 5(a) graphically presents the number of function evaluations per initial BAC obtained for the TRT004 case. Similarly to the CERR case, we can observe that the $\text{SD}_{\mathcal{N}_1}$ algorithm needs more #Evals than all the other algorithms considered in this study. The proposed rVNS algorithm, in the same way as for the CERR case, presents stable #Evals values. In particular, its rVNS_ND version needs fewer #Evals to solve the BAO problem, resulting in less computational time than its rVNS_SD counterpart.

Fig. 5(b) graphically presents the obtained objective function values, z, for each algorithm at each initial BAC for the TRT004 case. As we can see, the obtained results do not vary considerably among the initial BACs. Unlike for the CERR case, the standard deviation of the $\text{SD}_{\mathcal{N}_2}$ is very close to the obtained by the rVNS algorithms. It is interesting to notice that for the TRT004 case, the SD and the ND algorithms generally present poor results w.r.t. their z value when using the neighbourhood definition $\mathcal{N}_2$. This is different from what happened for the CERR case, where the poorest values are obtained when using the $\mathcal{N}_1$ neighbourhood definition. Thus, using both neighbourhood definitions, $\mathcal{N}_1$ and $\mathcal{N}_2$, at the same time, as the proposed rVNS algorithms do, seems to be an efficient strategy that allows us to maintain consistent results independently of the case and initial BAC that is considered.

Table III shows, again, the initial BAC, the number of iterations the algorithm needs before convergence (Iter.), the number of function evaluations (#Evals), and the best objective function (z) value obtained by the algorithm for the TRT004 case. Again, all these values are getting after 30 runs for each initial BAC. Thus, the reported values are the average of these 30 runs. The two algorithms with the best average performance regarding

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TABLE III
RESULTS OBTAINED FOR EACH ALGORITHM AND INITIAL BAC, FOR THE TRT004 INSTANCE

	rVNS_SD			rVNS_ND			$SD_{\mathcal{N}_1}$			$ND_{\mathcal{N}_1}$			$SD_{\mathcal{N}_2}$			$ND_{\mathcal{N}_2}$		
BAC	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z
1	6.50	65.00	0.2011	10.86	40.83	0.2013	13.00	130.00	**0.1990**	15.00	80.00	**0.1990**	3.30	**33.00**	0.2032	5.30	**23.93**	0.2034
2	6.60	66.00	**0.2016**	7.90	**33.90**	0.2018	9.00	90.00	**0.2003**	10.00	36.00	0.2022	3.80	38.33	0.2027	4.00	**24.40**	0.2038
3	7.53	75.33	**0.2007**	9.87	**36.17**	**0.2017**	9.00	90.00	0.2030	9.00	48.00	0.2030	4.03	40.33	0.2023	5.27	**23.70**	0.2027
4	6.27	62.67	0.2007	8.57	36.33	0.2009	9.00	90.00	**0.1999**	14.00	45.00	**0.2002**	3.50	**35.00**	0.2031	4.07	**20.90**	0.2033
5	7.67	76.67	**0.2015**	13.03	**43.67**	**0.2014**	16.00	160.00	0.2026	20.00	57.00	0.2017	4.83	48.33	0.2021	7.37	**27.53**	0.2031
6	6.73	67.33	0.2005	8.40	**33.50**	0.2008	11.00	110.00	**0.2000**	13.00	52.00	**0.2000**	3.83	38.33	0.2019	4.90	**23.50**	0.2029
7	6.00	60.00	**0.2010**	7.73	**31.13**	**0.2011**	8.00	80.00	0.2021	9.00	39.00	0.2014	3.53	35.33	0.2023	4.27	**23.10**	0.2026
8	7.13	71.33	0.2012	12.90	43.23	0.2014	22.00	220.00	**0.2009**	24.00	64.00	**0.1991**	4.13	**41.33**	0.2023	6.83	**27.433**	0.2041
9	7.20	72.00	**0.2009**	11.17	**40.13**	0.2013	16.00	160.00	**0.1996**	17.00	75.00	0.2010	4.47	44.67	0.2024	6.70	**27.17**	0.2025
10	9.93	99.33	**0.2014**	13.87	**47.73**	**0.2018**	12.00	120.00	0.2036	30.00	95.00	0.2027	5.60	56.00	0.2027	8.33	**33.60**	0.2025
$\bar{\mathbf{x}}$	**7.16**	**71.57**	**0.2011**	**10.43**	**38.66**	**0.2013**	**12.50**	**125.00**	**0.2011**	**16.10**	**59.10**	**0.2010**	**4.11**	**41.07**	**0.2025**	**5.71**	**25.53**	**0.2031**

TABLE IV
SUMMARY OF THE (AVERAGE) RESULTS OBTAINED FOR EACH ALGORITHM FOR THE ALL 7 INSTANCES CONSIDERED IN THIS STUDY

	rVNS_SD			rVNS_ND			$SD_{\mathcal{N}_1}$			$ND_{\mathcal{N}_1}$			$SD_{\mathcal{N}_2}$			$ND_{\mathcal{N}_2}$		
Case	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z
CERR	**8.76**	67.75	**0.6034**	12.80	**47.35**	**0.6005**	13.00	131.00	0.6114	17.98	61.80	0.6076	**4.24**	**42.78**	0.6059	20.16	61.34	0.6072
TRT003	8.22	82.22	**0.2194**	11.89	**42.63**	**0.2200**	15.5	155.00	0.2210	21.85	70.57	**0.2200**	**4.88**	**48.83**	0.2208	**6.63**	**27.50**	0.2213
TRT004	7.16	71.57	**0.2011**	10.43	**38.66**	0.2013	12.50	125.00	**0.2011**	16.10	59.10	**0.2010**	**4.11**	41.07	0.2025	**5.71**	**25.53**	0.2031
TRT007	8.73	87.30	0.2747	12.05	**43.14**	0.2769	20.40	204.00	**0.2703**	24.64	74.37	**0.2705**	**4.32**	43.23	0.2796	**6.05**	**25.89**	0.2804
TRT008	8.04	80.37	**0.2094**	10.92	**40.53**	0.2099	14.70	147.00	0.2101	19.29	67.55	**0.2095**	**4.83**	48.27	0.2104	**6.63**	**27.76**	0.2108
TRT009	7.49	74.90	**0.2543**	11.29	**41.16**	**0.2549**	12.00	120.00	0.2562	15.16	54.48	0.2563	**4.99**	49.87	0.2550	**7.24**	**30.04**	0.2553
TRT0010	7.96	79.60	**0.5754**	11.53	**42.70**	**0.5768**	14.70	147.00	0.5800	18.13	63.05	0.5794	**4.88**	48.83	**0.5768**	**7.28**	**29.91**	0.5775

#Evals and z have been highlighted. Again, the best z value is highlighted and underlined, while the second best z value is highlighted only. As for the CERR case, we can note that the exploitation strategy ($\mathcal{N}_1$) generally presents more evaluations than the exploration strategy ($\mathcal{N}_2$).

For each initial BAC, the proposeed the proposed rVNS algorithms obtain very low standard deviation values, with a standard deviation of 0.000367 for the rVNS_SD and 0.000334 for the rVNS_ND. While algorithms such as LS $SD_{\mathcal{N}_1}$, $ND_{\mathcal{N}_1}$, $SD_{\mathcal{N}_2}$ and $ND_{\mathcal{N}_2}$ have up to 4 times more standard deviation. In particular, the rVNS_SD obtains the best performance in 3 out of the 10 initial BACs (highlighted and underlined in Table III), being the second algorithm with the best performance. Further, the rVNS_SD obtains the second best z value in 4 out of the remaining 7 initial BACs (highlighted only in Table III). The rVNS_ND obtains the best performance in 2 out of the ten initial BACs (highlighted and underlined in Table III) and the second best results in 3 out of the remaining 8 initial BACs (highlighted only in Table III). The algorithm that exhibits the best performance regarding the obtained z value is the $SD_{\mathcal{N}_1}$, although this algorithm needs up to three times more #Evals than the proposed rVNS algorithms. In general, we can see that the rVNS algorithms are quite competitive w.r.t. the obtained objective function values, z, and faster than the best LS algorithm included in this study. This is important as we need to generate an acceptable treatment plan as faster as possible.

Finally, Table IV shows a summary of the results obtained for each instance considered in this study. As mentioned previously, the detailed results per instance can be seen in the Appendix section.

V. CONCLUSION

This paper presents a variant of the VNS framework, called reduced VNS, to solve radiation therapy's beam angle optimisation problem. Unlike traditional VNS algorithms, the proposed rVNS does not need any transition rule to move from one neighbourhood to the other, as all the neighbourhood definitions are used simultaneously at the same iteration. This strategy allows us to explore and exploit the search space simultaneously at each iteration of the algorithm.

We run our algorithm on 7 prostate cases, the first obtained from the literature (CERR) and the remaining 6 from a Chilean hospital. The proposed rVNS algorithm can find quality treatment plans compared to previously proposed LS algorithms. Furthermore, the rVNS algorithm converges faster than both the $SD_{\mathcal{N}_1}$ and the $SD_{\mathcal{N}_2}$ local search strategies, previously proposed in the literature. The rVNS strategy presented in this work exhibits stable results for different initial BACs needing fewer function evaluations compared to the LS algorithms.

The strategy of simultaneously exploring and exploiting the search space allows the proposed rVNS algorithm to escape from local minima, obtaining competitive results. Unlike the traditional LS algorithms, the proposed rVNS algorithm maintains consistent results independently of the case and the initial BAC that is considered. One of the challenges of the BAO problem is that the search space is too large. Strategies such as the rVNS algorithms presented in this study allow us to better explore this search space obtaining high-quality treatment plans. However, other strategies might be considered. In future work, we expect to implement strategies to reduce the search space by using ad-hoc rules. These reduction strategies should allow the rVNS algorithms to find better treatment plans and/or converge even faster. We are also interested in extending our proposed rVNS approach to multi-objective optimisation.

APPENDIX

A Statistical Analysis: Resume P-Values

B. Results for the TRT00X Cases

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TABLE V
SUMMARY OF THE RESULTS OBTAINED AFTER APPLYING THE WILCOXON-MANN-WHITNEY'S FOR THE ALL 7 INSTANCES INCLUDED IN THIS STUDY

(a) Wilcoxon-Mann-Whitney's test for the CERR instance

p-value	rVNS_SD	rVNS_ND	$SD_{\mathcal{N}_1}$	$ND_{\mathcal{N}_1}$	$SD_{\mathcal{N}_2}$	$ND_{\mathcal{N}_2}$
rVNS_SD	-	4	1	6	5	7
rVNS_ND	0	-	1	5	8	8
$SD_{\mathcal{N}_1}$	0	0	-	3	4	5
$ND_{\mathcal{N}_1}$	3	4	5	-	6	8
$SD_{\mathcal{N}_2}$	0	0	0	1	-	4
$ND_{\mathcal{N}_2}$	0	0	0	2	0	-

(b) Wilcoxon-Mann-Whitney's test for the TRT003 instance

p-value	rVNS_SD	rVNS_ND	$SD_{\mathcal{N}_1}$	$ND_{\mathcal{N}_1}$	$SD_{\mathcal{N}_2}$	$ND_{\mathcal{N}_2}$
rVNS_SD	-	4	1	3	7	8
rVNS_ND	0	-	1	3	5	8
$SD_{\mathcal{N}_1}$	0	0	-	0	0	0
$ND_{\mathcal{N}_1}$	2	4	0	-	6	8
$SD_{\mathcal{N}_2}$	0	0	0	2	-	2
$ND_{\mathcal{N}_2}$	0	0	0	2	0	-

(c) Wilcoxon-Mann-Whitney's test for the TRT004 instance

p-value	rVNS_SD	rVNS_ND	$SD_{\mathcal{N}_1}$	$ND_{\mathcal{N}_1}$	$SD_{\mathcal{N}_2}$	$ND_{\mathcal{N}_2}$
rVNS_SD	-	1	0	6	10	10
rVNS_ND	0	-	0	4	10	10
$SD_{\mathcal{N}_1}$	0	0	-	3	4	5
$ND_{\mathcal{N}_1}$	3	4	5	-	6	8
$SD_{\mathcal{N}_2}$	0	0	0	1	-	4
$ND_{\mathcal{N}_2}$	0	0	0	0	0	-

(d) Wilcoxon-Mann-Whitney's test for the TRT007 instance

p-value	rVNS_SD	rVNS_ND	$SD_{\mathcal{N}_1}$	$ND_{\mathcal{N}_1}$	$SD_{\mathcal{N}_2}$	$ND_{\mathcal{N}_2}$
rVNS_SD	-	4	0	0	10	10
rVNS_ND	0	-	0	0	8	9
$SD_{\mathcal{N}_1}$	0	1	-	2	7	7
$ND_{\mathcal{N}_1}$	10	10	1	-	10	10
$SD_{\mathcal{N}_2}$	0	0	0	0	-	2
$ND_{\mathcal{N}_2}$	0	0	0	0	0	-

(e) Wilcoxon-Mann-Whitney's test for the TRT008 instance

p-value	rVNS_SD	rVNS_ND	$SD_{\mathcal{N}_1}$	$ND_{\mathcal{N}_1}$	$SD_{\mathcal{N}_2}$	$ND_{\mathcal{N}_2}$
rVNS_SD	-	4	0	5	8	10
rVNS_ND	1	-	0	4	4	6
$SD_{\mathcal{N}_1}$	0	0	-	0	0	0
$ND_{\mathcal{N}_1}$	3	4	0	-	6	7
$SD_{\mathcal{N}_2}$	0	0	0	0	-	1
$ND_{\mathcal{N}_2}$	0	0	0	0	0	-

(f) Wilcoxon-Mann-Whitney's test for the TRT009 instance

p-value	rVNS_SD	rVNS_ND	$SD_{\mathcal{N}_1}$	$ND_{\mathcal{N}_1}$	$SD_{\mathcal{N}_2}$	$ND_{\mathcal{N}_2}$
rVNS_SD	-	5	0	7	7	9
rVNS_ND	0	-	1	5	2	5
$SD_{\mathcal{N}_1}$	0	0	-	0	0	0
$ND_{\mathcal{N}_1}$	0	1	0	-	2	4
$SD_{\mathcal{N}_2}$	0	1	3	5	-	2
$ND_{\mathcal{N}_2}$	0	0	2	5	0	-

(g) Wilcoxon-Mann-Whitney's test for the TRT010 instance

p-value	rVNS_SD	rVNS_ND	$SD_{\mathcal{N}_1}$	$ND_{\mathcal{N}_1}$	$SD_{\mathcal{N}_2}$	$ND_{\mathcal{N}_2}$
rVNS_SD	-	6	3	9	7	8
rVNS_ND	0	-	1	6	2	4
$SD_{\mathcal{N}_1}$	0	0	-	0	0	0
$ND_{\mathcal{N}_1}$	0	1	1	-	1	2
$SD_{\mathcal{N}_2}$	0	1	1	7	-	3
$ND_{\mathcal{N}_2}$	0	0	1	6	0	-

TABLE VI
RESULTS OBTAINED FOR EACH ALGORITHM AND INITIAL BAC, FOR THE TRT003 INSTANCE

	rVNS_SD			rVNS_ND			$SD_{\mathcal{N}_1}$			$ND_{\mathcal{N}_1}$			$SD_{\mathcal{N}_2}$			$ND_{\mathcal{N}_2}$		
BAC	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z
1	6.8	68	**0.2199**	12.17	**43.07**	**0.2199**	11	110	0.2236	17.83	57.3	0.2217	5.73	57.33	**0.2203**	6.27	**26.97**	0.2210
2	7.03	70.3	**0.2196**	10	40.23	0.2202	14	140	**0.2184**	15.97	59.37	0.2197	3.67	**36.67**	0.2214	3.5	**20.6**	0.2220
3	8.13	81.33	**0.2187**	10.47	**36.73**	0.2200	10	100	0.2206	20.33	68.6	**0.2190**	4.47	44.67	0.2207	5.63	**25.83**	0.2213
4	8.53	85.33	**0.2185**	11.5	42.3	0.2195	17	170	0.2188	21.97	70.9	**0.2180**	3.9	**39**	0.2207	6.83	**28.2**	0.2210
5	11.53	115.33	**0.2183**	16.03	**52.93**	0.2194	32	320	**0.2185**	34.13	92.73	0.2188	5.97	59.67	0.2210	9.47	**32.17**	0.2218
6	8.87	88.67	**0.2209**	11.83	**42.23**	**0.2200**	11	110	0.2257	19.90	72	0.2240	6.03	60.33	0.2212	8.33	**35.80**	**0.2209**
7	6.6	66.00	**0.2207**	6.73	**30.47**	0.2217	6	60	0.2223	11.07	44.4	**0.2212**	3.67	36.67	0.2220	3.90	**23.03**	0.2222
8	8.4	84	**0.2189**	14.37	**48.70**	0.2204	20	200	0.2234	29.4	90	**0.2200**	5.47	54.67	**0.2200**	8.27	**28.97**	0.2210
9	8.1	81	**0.2189**	12.27	**43.00**	0.2198	14	140	0.2192	20.17	68.7	**0.2191**	4.87	48.67	0.2205	7.1	**26.97**	0.2210
10	9.03	90.33	**0.2192**	13.50	46.63	0.2198	20	200	0.2193	27.73	81.7	**0.2183**	5.07	50.67	0.2200	6.97	26.43	0.2208
$\bar{\mathbf{x}}$	**8.22**	**82.22**	**0.2194**	**11.89**	**42.63**	**0.2200**	**15.5**	**155.00**	**0.2210**	**21.85**	**70.57**	**0.2200**	**4.88**	**48.83**	**0.2208**	**6.63**	**27.50**	**0.2213**

TABLE VII
RESULTS OBTAINED FOR EACH ALGORITHM AND INITIAL BAC, FOR THE TRT007 INSTANCE

	rVNS_SD			rVNS_ND			$SD_{\mathcal{N}_1}$			$ND_{\mathcal{N}_1}$			$SD_{\mathcal{N}_2}$			$ND_{\mathcal{N}_2}$		
BAC	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z
1	9.33	93.33	0.2760	12.97	48.50	0.2786	20.00	200.00	**0.2698**	21.53	63.27	**0.2701**	3.83	**38.33**	0.2804	5.83	**24.50**	0.2851
2	7.27	72.67	**0.2737**	6.90	30.57	0.2757	11.00	110.00	**0.2685**	14.20	52.27	**0.2685**	3.00	**30.00**	0.2798	2.80	**18.57**	0.2814
3	9.50	95.00	0.2757	11.53	**40.63**	0.2804	20.00	200.00	**0.2685**	22.13	63.23	**0.2690**	5.03	50.33	0.2800	7.70	**32.40**	0.2790
4	7.63	76.33	0.2747	10.23	39.67	0.2762	21.00	210.00	**0.2701**	24.80	81.90	**0.2713**	3.83	**38.33**	0.2790	4.60	**22.43**	0.2803
5	12.37	123.67	0.2735	18.20	**56.73**	0.2746	28.00	280.00	**0.2710**	33.13	85.40	**0.2700**	6.20	62.00	0.2801	9.83	**32.73**	0.2806
6	8.23	82.33	0.2762	13.80	46.80	0.2772	26.00	260.00	**0.2726**	31.37	97.60	**0.2733**	4.33	**43.33**	0.2815	6.20	**27.70**	0.2802
7	5.90	59.00	0.2750	7.17	32.53	0.2753	9.00	90.00	**0.2712**	11.93	45.60	**0.2713**	2.13	**21.33**	0.2805	2.57	**16.73**	0.2806
8	8.80	88.00	0.2757	14.30	**48.27**	0.2757	31.00	310.00	**0.2710**	35.37	101.70	**0.2708**	5.00	50.00	0.2801	7.37	**28.77**	0.2779
9	8.57	85.67	0.2717	10.87	**39.70**	0.2763	16.00	160.00	**0.2702**	25.33	75.90	**0.2701**	4.47	44.67	0.2756	5.60	**24.30**	0.2772
10	9.70	97.00	0.2749	14.50	**47.97**	0.2787	22.00	220.00	**0.2701**	26.60	76.80	**0.2706**	5.40	54.00	0.2792	7.97	**30.77**	0.2818
$\bar{\mathbf{x}}$	**8.73**	**87.30**	**0.2747**	**12.05**	**43.14**	**0.2769**	**20.40**	**204.00**	**0.2703**	**24.64**	**74.37**	**0.2705**	**4.32**	**43.23**	**0.2796**	**6.05**	**25.89**	**0.2804**

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TABLE VIII
RESULTS OBTAINED FOR EACH ALGORITHM AND INITIAL BAC, FOR THE TRT008 INSTANCE

	rVNS_SD			rVNS_ND			$SD_{\mathcal{N}_1}$			$ND_{\mathcal{N}_1}$			$SD_{\mathcal{N}_2}$			$ND_{\mathcal{N}_2}$		
BAC	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z
1	8.73	87.33	**0.2090**	11.03	**42.40**	0.2101	14.00	140.00	**0.2094**	16.37	59.33	0.2099	4.93	49.33	0.2103	7.07	**29.67**	0.2106
2	6.50	65.00	**0.2090**	5.40	**28.40**	0.2104	12.00	120.00	**0.2078**	13.93	64.13	**0.2078**	3.80	38.00	0.2100	4.77	**24.60**	0.2104
3	7.17	71.67	**0.2098**	12.93	**45.53**	**0.2091**	12.00	120.00	0.2111	17.40	52.37	0.2103	4.57	45.67	0.2105	6.70	**27.20**	0.2107
4	7.83	78.33	**0.2099**	11.03	**42.37**	**0.2094**	8.00	80.00	0.2134	14.57	61.87	0.2110	5.63	56.33	0.2110	5.77	**25.70**	0.2119
5	9.13	91.33	**0.2089**	11.87	**41.47**	0.2097	16.00	160.00	0.2099	30.10	89.57	**0.2079**	5.53	55.33	0.2093	8.97	**31.67**	0.2098
6	8.50	85.00	**0.2094**	13.03	**44.43**	0.2100	24.00	240.00	**0.2093**	23.97	85.67	0.2097	5.67	56.67	0.2101	8.00	**31.53**	0.2109
7	6.63	66.33	0.2098	7.07	33.93	0.2107	12.00	120.00	**0.2082**	12.97	50.20	**0.2085**	3.20	**32.00**	0.2114	3.73	**22.33**	0.2116
8	9.27	92.67	**0.2095**	12.10	**42.30**	**0.2098**	19.00	190.00	0.2123	24.57	84.00	0.2112	5.20	52.00	0.2102	7.17	**28.50**	0.2107
9	7.13	71.33	**0.2092**	11.53	42.30	**0.2095**	8.00	80.00	0.2102	12.97	43.97	0.2101	3.90	**39.00**	0.2107	6.23	**26.80**	0.2107
10	9.47	94.67	**0.2092**	13.20	**42.17**	0.2104	22.00	220.00	0.2094	26.10	84.37	**0.2090**	5.83	58.33	0.2101	7.93	**29.60**	0.2106
$\bar{x}$	**8.04**	**80.37**	**0.2094**	**10.92**	**40.53**	**0.2099**	**14.70**	**147.00**	**0.2101**	**19.29**	**67.55**	**0.2095**	**4.83**	**48.27**	**0.2104**	**6.63**	**27.76**	**0.2108**

TABLE IX
RESULTS OBTAINED FOR EACH ALGORITHM AND INITIAL BAC, FOR THE TRT009 INSTANCE

	rVNS_SD			rVNS_ND			$SD_{\mathcal{N}_1}$			$ND_{\mathcal{N}_1}$			$SD_{\mathcal{N}_2}$			$ND_{\mathcal{N}_2}$		
BAC	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z
1	6.47	64.67	**0.2539**	10.07	41.13	**0.2543**	10.00	100.00	0.2547	12.67	46.17	0.2551	3.87	**38.67**	0.2547	6.30	**28.17**	0.2548
2	4.90	49.00	**0.2548**	6.73	**30.87**	**0.2547**	6.00	60.00	**0.2547**	7.57	32.50	**0.2547**	4.40	44.00	0.2552	4.63	**25.17**	0.2554
3	7.67	76.67	**0.2549**	12.93	43.97	**0.2556**	9.00	90.00	0.2604	9.23	**39.50**	0.2605	5.67	56.67	0.2557	8.13	**32.07**	0.2557
4	8.53	85.33	**0.2537**	10.70	43.43	0.2551	5.00	50.00	0.2596	6.93	**29.57**	0.2591	5.40	54.00	**0.2546**	7.23	**31.23**	0.2555
5	8.70	87.00	0.2546	12.67	**42.67**	0.2550	19.00	190.00	**0.2543**	26.63	92.30	**0.2544**	4.87	48.67	0.2554	7.80	**30.67**	0.2560
6	8.80	88.00	**0.2541**	11.90	**41.73**	**0.2545**	10.00	100.00	0.2595	11.17	43.37	0.2598	5.23	52.33	0.2550	8.43	**32.00**	0.2550
7	5.77	57.67	0.2544	9.87	**38.83**	0.2545	10.00	100.00	**0.2539**	13.30	49.00	**0.2540**	4.50	45.00	0.2547	6.60	**29.67**	0.2550
8	8.40	84.00	**0.2538**	12.77	**43.07**	0.2549	16.00	160.00	0.2546	23.40	68.60	**0.2541**	4.97	49.67	0.2547	7.97	**31.87**	0.2552
9	7.93	79.33	**0.2545**	11.77	**39.90**	0.2556	18.00	180.00	0.2561	19.70	66.90	0.2560	5.63	56.33	**0.2547**	7.83	**30.37**	0.2551
10	7.73	77.33	**0.2539**	13.53	45.97	**0.2544**	17.00	170.00	0.2545	21.00	76.87	0.2548	5.33	53.33	0.2550	7.43	29.20	0.2553
$\bar{x}$	**7.49**	**74.90**	**0.2543**	**11.29**	**41.16**	**0.2549**	**12.00**	**120.00**	**0.2562**	**15.16**	**54.48**	**0.2563**	**4.99**	**49.87**	**0.2550**	**7.24**	**30.04**	**0.2553**

TABLE X
RESULTS OBTAINED FOR EACH ALGORITHM AND INITIAL BAC, FOR THE TRT0010 INSTANCE

	rVNS_SD			rVNS_ND			$SD_{\mathcal{N}_1}$			$ND_{\mathcal{N}_1}$			$SD_{\mathcal{N}_2}$			$ND_{\mathcal{N}_2}$		
BAC	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z	Iter.	#Evals.	z
1	8.77	87.67	**0.5755**	11.87	**43.87**	**0.5760**	18.00	180.00	0.5818	21.17	76.33	0.5804	4.83	48.33	0.5769	6.40	**27.40**	0.5778
2	5.67	56.67	**0.5762**	5.20	28.60	0.5786	3.00	30.00	0.5800	5.00	27.87	0.5798	3.37	33.67	**0.5770**	3.20	20.20	0.5788
3	6.40	64.00	**0.5773**	10.40	**39.63**	**0.5768**	11.00	110.00	0.5791	13.47	45.03	0.5788	4.90	49.00	**0.5773**	6.57	**28.30**	0.5775
4	7.47	74.67	**0.5759**	10.63	**41.63**	**0.5755**	14.00	140.00	0.5769	16.93	68.77	0.5769	4.63	46.33	0.5768	6.63	**29.77**	0.5773
5	9.93	99.33	**0.5740**	15.03	50.00	0.5776	25.00	250.00	**0.5763**	31.83	93.33	0.5776	5.70	57.00	0.5765	9.50	32.80	0.5790
6	9.67	96.67	**0.5757**	15.17	**52.27**	0.5775	21.00	210.00	0.5846	22.80	76.43	0.5859	5.93	59.33	0.5780	11.30	**41.80**	**0.5769**
7	7.30	73.00	**0.5753**	8.93	37.40	0.5781	5.00	50.00	0.5845	6.30	**30.50**	0.5836	4.50	45.00	**0.5765**	6.13	**29.40**	0.5768
8	8.47	84.67	0.5758	14.13	**46.40**	0.5765	25.00	250.00	**0.5731**	24.37	82.30	**0.5749**	5.47	54.67	0.5766	9.40	**35.20**	0.5766
9	7.07	70.67	**0.5740**	9.53	39.23	**0.5750**	5.00	50.00	0.5803	7.37	**35.43**	0.5803	4.13	41.33	0.5754	5.40	**24.67**	0.5771
10	8.87	88.67	**0.5747**	14.40	**48.00**	0.5766	20.00	200.00	0.5832	32.03	94.53	0.5757	5.37	53.67	**0.5765**	8.30	**29.60**	0.5777
$\bar{x}$	**7.96**	**79.60**	**0.5754**	**11.53**	**42.70**	**0.5768**	**14.70**	**147.00**	**0.5800**	**18.13**	**63.05**	**0.5794**	**4.88**	**48.83**	**0.5768**	**7.28**	**29.91**	**0.5775**

REFERENCES

[1] M. Ehrgott and R. Johnston, "Optimisation of beam directions in intensity modulated radiation therapy planning," *OR Spectr.*, vol. 25, no. 2, pp. 251–264, 2003.

[2] G. Cabrera et al., "A matheuristic approach to solve the multiobjective beam angle optimization problem in intensity-modulated radiation therapy," *Int. Trans. Oper. Res.*, vol. 25, no. 1, pp. 243–268, 2018.

[3] A. Pugachev, A. Boyer, and L. Xing, "Beam orientation optimization in intensity-modulated radiation treatment planning," *Med. Phys.*, vol. 27, no. 6, pp. 1238–1245, 2000.

[4] S. Mohammadi, C. Shang, Z. Ouhib, T. Leventouri, and G. Kalantzis, "A computational study on different penalty approaches for constrained optimization in radiation therapy treatment planning with a simulated annealing algorithm," in *Proc. IEEE/ACIS 16th Int. Conf. Softw. Eng., Artif. Intell., Netw. Parallel/Distrib. Comput.*, 2015, pp. 1–6.

[5] J. Dias, H. Rocha, B. Ferreira, and M. do Carmo Lopes, "A genetic algorithm with neural network fitness function evaluation for IMRT beam angle optimization," *Central Eur. J. Operations Res.*, vol. 22, no. 3, pp. 431–455, 2014.

[6] Y. Li and D. Yao, "Accelerating the radiotherapy planning with a hybrid method of genetic algorithm and ant colony system," in *Proc. Int. Conf. Natural Comput.*, 2006, pp. 340–349.

[7] P. Carrasqueira et al., "An automated bi-level optimization approach for IMRT," *Int. Trans. Oper. Res.*, vol. 30, no. 1, pp. 224–238, 2021.

[8] H. Yarmand and D. Craft, "Two effective heuristics for beam angle optimization in radiation therapy," *Simulation*, vol. 94, no. 11, pp. 1041–1049, 2018, doi: 10.1177/0037549718761108.

[9] A. Ghanbarzadeh, M. Pouladian, A. Shabestani Monfared, and S. R. Mahdavi, "The scatter search based algorithm for beam angle optimization in intensity-modulated radiation therapy," *Comput. Math. Methods Med.*, vol. 2018, 2018, Art. no. 4571801.

[10] H. Yarmand and D. Craft, "Effective heuristics for beam angle optimization in radiation therapy," *Simulation*, vol. 94, no. 11, pp. 1041–1049, Nov. 2018.

[11] N. Ramar et al., "Objective function based ranking method for selection of optimal beam angles in IMRT," *Physica Medica*, vol. 69, pp. 44–51, Jan. 2020.

[12] S. Haseai, H. Arimura, K. Asai, T. Yoshitake, and Y. Shioyama, "Similar-cases-based planning approaches with beam angle optimizations using water equivalent path length for lung stereotactic body radiation therapy," *Radiological Phys. Technol.*, vol. 13, no. 2, pp. 119–127, Jun. 2020.

[13] V. T. Taasti, L. Hong, J. S. Shim, J. O. Deasy, and M. Zarepisheh, "Automating proton treatment planning with beam angle selection using Bayesian optimization," *Med. Phys.*, vol. 47, no. 8, pp. 3286–3296, 2020. [Online]. Available: https://aapm.onlinelibrary.wiley.com/doi/abs/10.1002/mp.14215

[14] R. Kaderka et al., "Toward automatic beam angle selection for pencil-beam scanning proton liver treatments: A deep learning-based approach," *Med. Phys.*, vol. 49, no. 7, pp. 4293–4304, 2022.

[15] W. Cheon et al., "Beam angle optimization for double-scattering proton delivery technique using an Eclipse application programming interface and convolutional neural network," *Front. Oncol.*, vol. 11, 2021, Art. no. 707464.

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[16] M. Field, N. Hardcastle, M. Jameson, N. Aherne, and L. Holloway, "Machine learning applications in radiation oncology," *Phys. Imag. Radiat. Oncol.*, vol. 19, pp. 13–24, 2021.

[17] G. Cabrera-Guerrero, N. Rodriguez, C. Lagos, E. Cabrera, and F. Johnson, "Local search algorithms for the beam angles' selection problem in radiotherapy," *Math. Problems Eng.*, vol. 2018, no. 1, pp. 1–9, 2018.

[18] G. Cabrera-Guerrero, C. Lagos, E. Cabrera, F. Johnson, J. M. Rubio, and F. Paredes, "Comparing local search algorithms for the beam angles selection in radiotherapy," *IEEE Access*, vol. 6, pp. 23701–23710, 2018.

[19] D. M. Aleman, A. Kumar, R. K. Ahuja, H. E. Romeijn, and J. F. Dempsey, "Neighborhood search approaches to beam orientation optimization in intensity modulated radiation therapy treatment planning," *J. Glob. Optim.*, vol. 42, no. 4, pp. 587–607, 2008.

[20] M. Gutierrez and G. Cabrera-Guerrero, "A variable neighbourhood search algorithm for the beam angle selection problem in radiation therapy," in *Proc. IEEE 37th Int. Conf. Chilean Comput. Sci. Soc.*, 2018, pp. 1–6.

[21] J. C. de Freitas, H. de Oliveira Florentino, A dos Santos Benedito, and D. R. Cantane, "Optimization model applied to radiotherapy planning problem with dose intensity and beam choice," *Appl. Math. Comput.*, vol. 387, 2020, Art. no. 124786.

[22] Qiuwen Wu, R. Mohan, and A. Niemierko, "IMRT optimization based on the generalized equivalent uniform dose (EUD)," in *Proc. IEEE 22nd Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2000, pp. 710–713.

[23] A. Niemierko, "Reporting and analyzing dose distributions: A concept of equivalent uniform dose," *Med. Phys.*, vol. 24, no. 1, pp. 103–113, 1997.

[24] A. Olafsson, R. Jeraj, and S. J. Wright, "Optimization of intensity-modulated radiation therapy with biological objectives," *Phys. Med. Biol.*, vol. 50, no. 22, 2005, Art. no. 5357.

[25] E. Thomas, O. Chapet, M. L. Kessler, T. S. Lawrence, and R. K. Ten Haken, "Benefit of using biologic parameters (EUD and NTCP) in IMRT optimization for treatment of intrahepatic tumors," *Int. J. Radiat. Oncol. Biol. Phys.*, vol. 62, no. 2, pp. 571–578, 2005.

[26] P. Hansen, N. Mladenović, and J. A. M. Pérez, "Variable neighbourhood search: Methods and applications," *Ann. Operations Res.*, vol. 175, no. 1, pp. 367–407, 2010.

[27] P. Hansen, N. Mladenović, R. Todosijević, and S. Hanafi, "Variable neighborhood search: Basics and variants," *EURO J. Comput. Optim.*, vol. 5, no. 3, pp. 423–454, 2017.

[28] G. Cabrera-Guerrero, A. Mason, A. Raith, and M. Ehrgott, "Pareto local search algorithms for the multi-objective beam angle optimisation problem," *J. Heuristics*, vol. 24, no. 2, pp. 205–238, 2018.

[29] J. O. Deasy, A. I. Blanco, and V. H. Clark, "CERR: A computational environment for radiotherapy research," *Med. Phys.*, vol. 30, no. 5, pp. 979–985, Apr. 2003. [Online]. Available: https://doi.org/10.1118/1.1568978

[30] M. Ehrgott, Ç. Güler, H. W. Hamacher, and L. Shao, "Mathematical optimization in intensity modulated radiation therapy," *Ann. Operations Res.*, vol. 175, no. 1, pp. 309–365, 2010.

[31] R. H. Byrd, J. Nocedal, and R. A. Waltz, "K nitro: An integrated package for nonlinear optimization," in *Large-Scale Nonlinear Optimization.*Berlin, Germany: Springer, 2006, pp. 35–59.

[32] W. L. Hays and R. L. Winkler, *Statistics: Probability, Inference, and Decision*. New York, NY, USA: Holt, Rinehart, and Winston, 1971.

[33] H. W. Lilliefors, "On the Kolmogorov-Smirnov test for normality with mean and variance unknown," *J. Amer. Stat. Assoc.*, vol. 62, no. 318, pp. 399–402, 1967.

[34] H. B. Mann and D. R. Whitney, "On a test of whether one of two random variables is stochastically larger than the other," *Ann. Math. Statist.*, vol. 18, no. 1, pp. 50–60, 1947.

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